

ROC CURVE



1. ROC空間將偽陽性率（FPR）為 X 軸，真陽性率（TPR）為 Y 軸
2. 給定一個二元分類模型和它的閾值，就能從所有樣本的（陽性/陰性）真實值和預測值計算出一個（ $X=FPR, Y=TPR$ ）座標點
3. 決策的閾值，也就是機率門檻值將會不同，ROC 曲線的目標是要呈現不同閾值會造成什麼不同的決策品質

	真實值	
預測值	TP	FP
	FN	TN

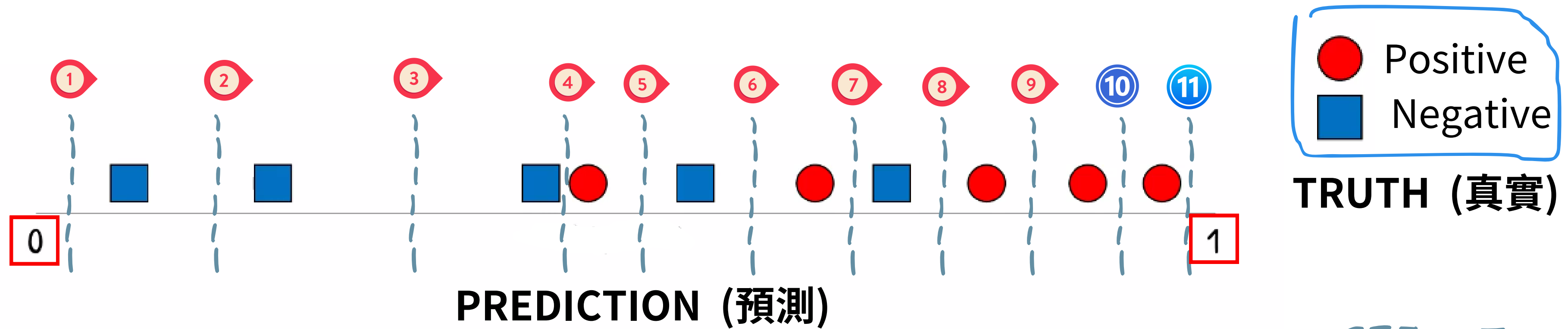
真陽性率 (True Positive Rate, TPR) =
真陽性數量 / 真實答案為正樣本數量

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

偽陽性率 (False Positive Rate, FPR) =
偽陽性數量 / 真實答案為負樣本數量

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

	真實值	
預測值	TP	FP
	FN	TN



←
接近0，代表"Negative"

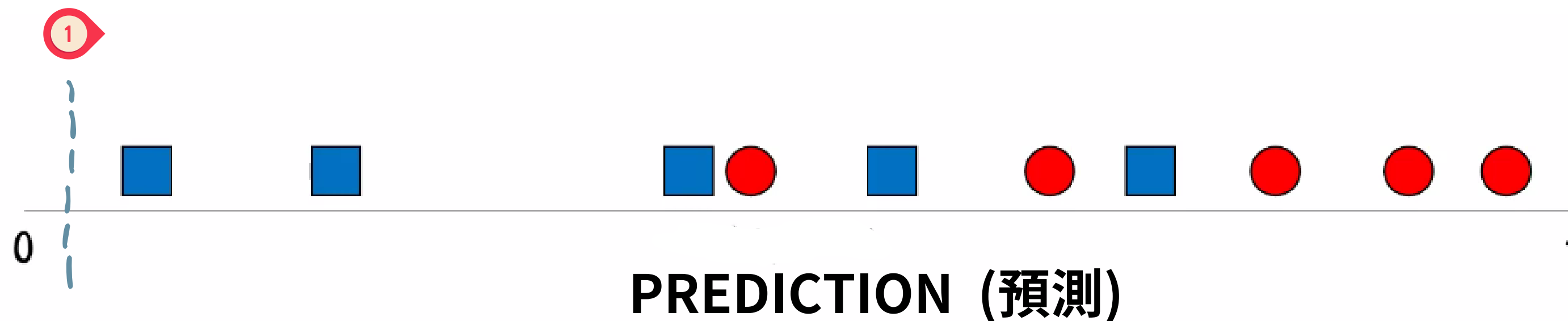
→
接近1，代表"Positive"

閾值 => 可以設定不同的數字

=> 不一定是0.5

=> 閾值設越大，為了對Negative的資料進行分類

=> 閾值設越小，為了對Positive的資料進行分類



1 閾值

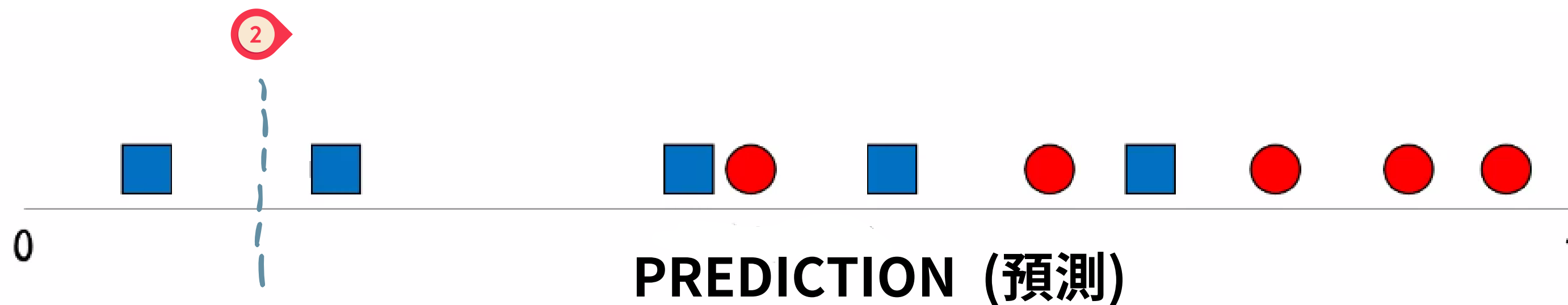
	真實值	
預測值	5	5
	0	0

$$TPR = TP / (TP + FN)$$

$$= 5 / (5 + 0) = \mathbf{1}$$

$$FPR = FP / (FP + TN)$$

$$= 5 / (5 + 0) = \mathbf{1}$$



2 閾值

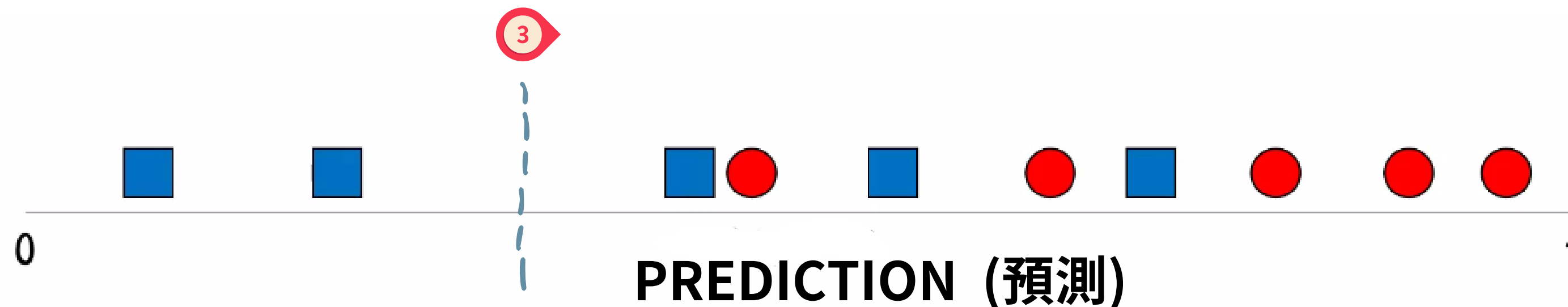
	真實值	
預測值	5	4
	0	1

$$TPR = TP / (TP + FN)$$

$$= 5 / (5 + 0) = 1$$

$$FPR = FP / (FP + TN)$$

$$= 4 / (4 + 1) = 0.8$$



● Positive
■ Negative

TRUTH (真實)

Cut-off(閾值)

3 閾值

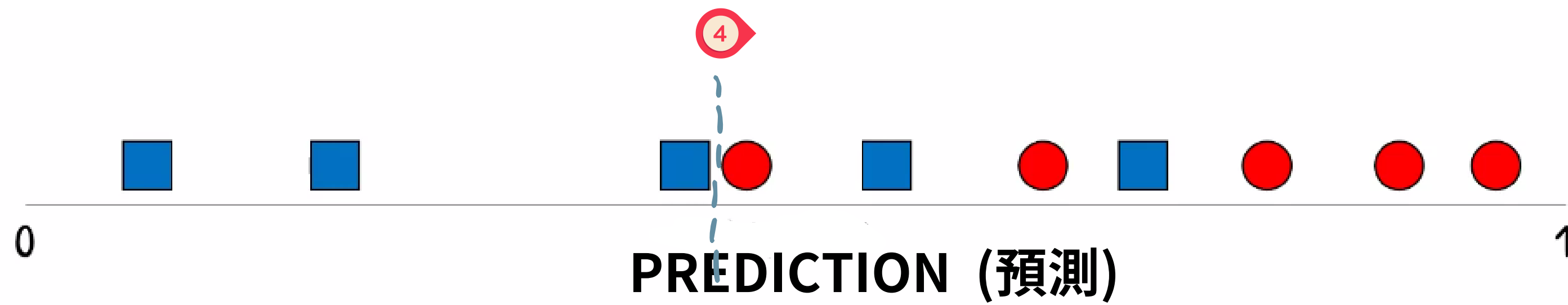
	真實值	
預測值	5	3
	0	2

$$TPR = TP / (TP + FN)$$

$$= 5 / (5 + 0) = \textcolor{red}{1}$$

$$FPR = FP / (FP + TN)$$

$$= 3 / (3 + 2) = \textcolor{red}{0.6}$$



Positive

Negative

TRUTH (真實)

Cut-off(閾值)

4

閾值

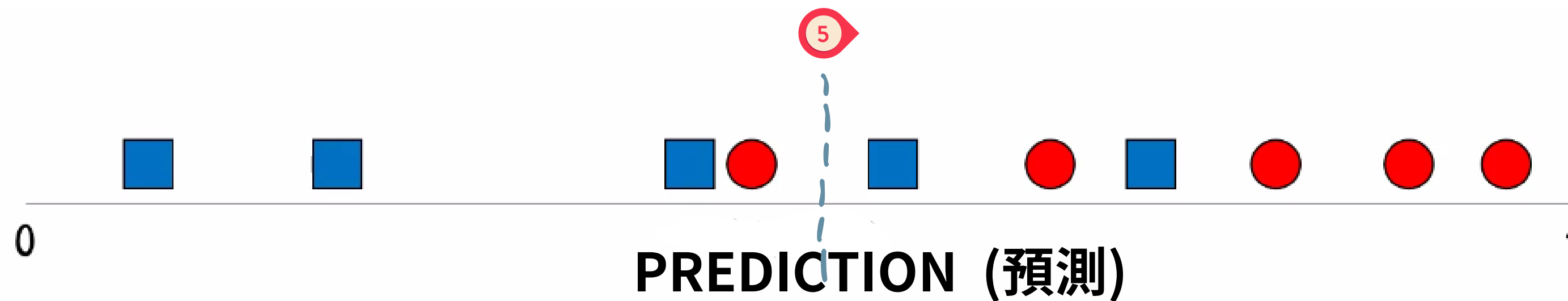
	真實值	
預測值	5	2
	0	3

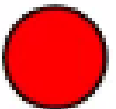
$$TPR = TP / (TP + FN)$$

$$= 5 / (5 + 0) = 1$$

$$FPR = FP / (FP + TN)$$

$$= 2 / (2 + 3) = 0.4$$



 Positive
 Negative

TRUTH (真實)

Cut-off(閾值)

5 閾值

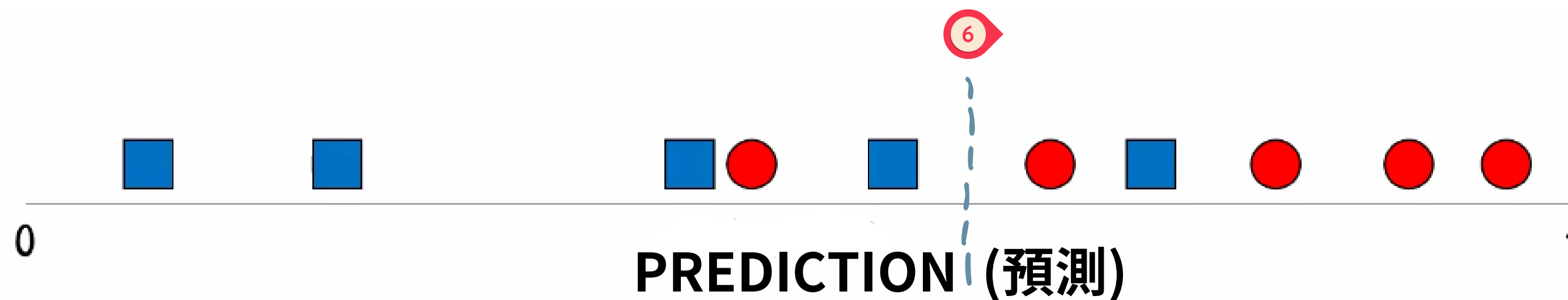
	真實值	
預測值	4	2
	1	3

$$TPR = TP / (TP + FN)$$

$$= 4 / (4 + 1) = 0.8$$

$$FPR = FP / (FP + TN)$$

$$= 2 / (2 + 3) = 0.4$$



 Positive
 Negative

TRUTH (真實)

Cut-off(閾值)

6 閾值

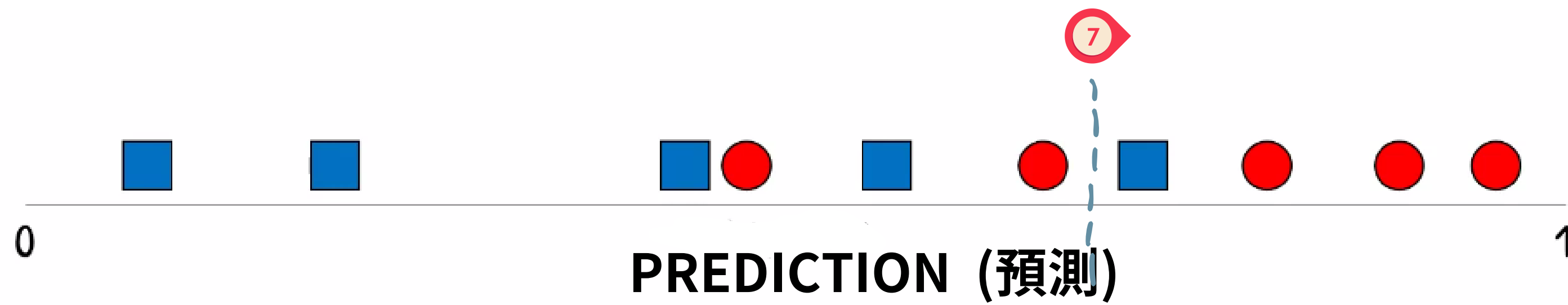
	真實值	
預測值	4	1
	1	4

$$TPR = TP / (TP + FN)$$

$$= 4 / (4 + 1) = 0.8$$

$$FPR = FP / (FP + TN)$$

$$= 1 / (1 + 4) = 0.2$$



Positive

Negative

TRUTH (真實)

Cut-off(閾值)

7

閾值

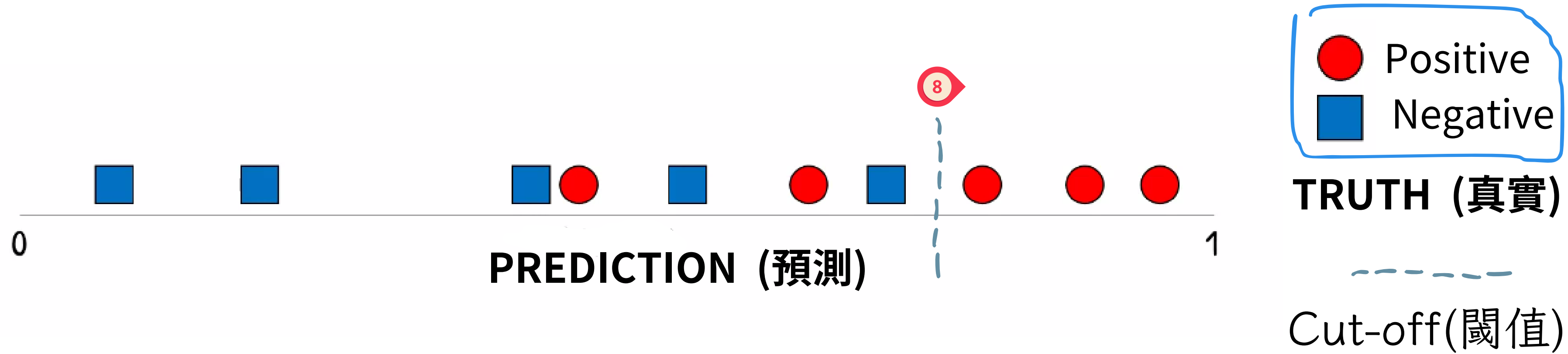
	真實值	
預測值	3	1
	2	4

$$TPR = TP / (TP + FN)$$

$$= 3 / (3 + 2) = 0.6$$

$$FPR = FP / (FP + TN)$$

$$= 1 / (1 + 4) = 0.2$$



8 閾值

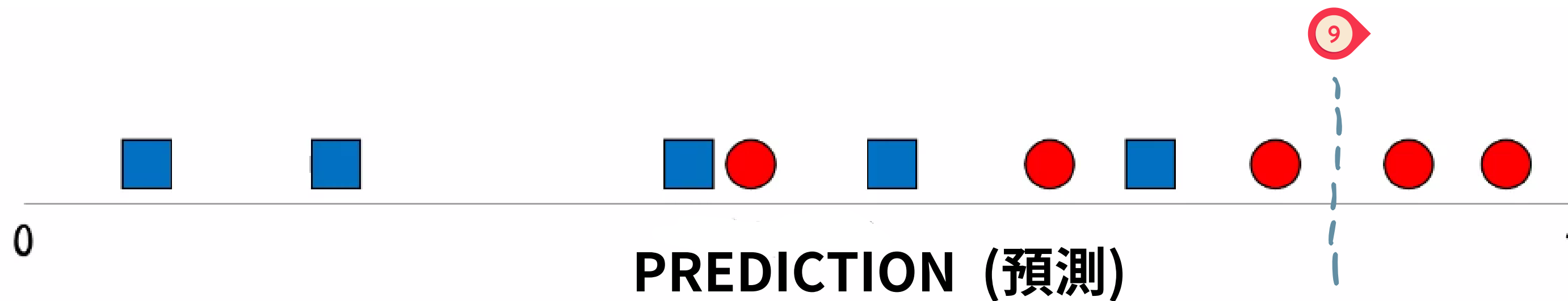
	真實值	
預測值	3	0
	2	5

$$TPR = TP / (TP + FN)$$

$$= 3 / (3 + 2) = 0.6$$

$$FPR = FP / (FP + TN)$$

$$= 0 / (0 + 5) = 0$$



9 閾值

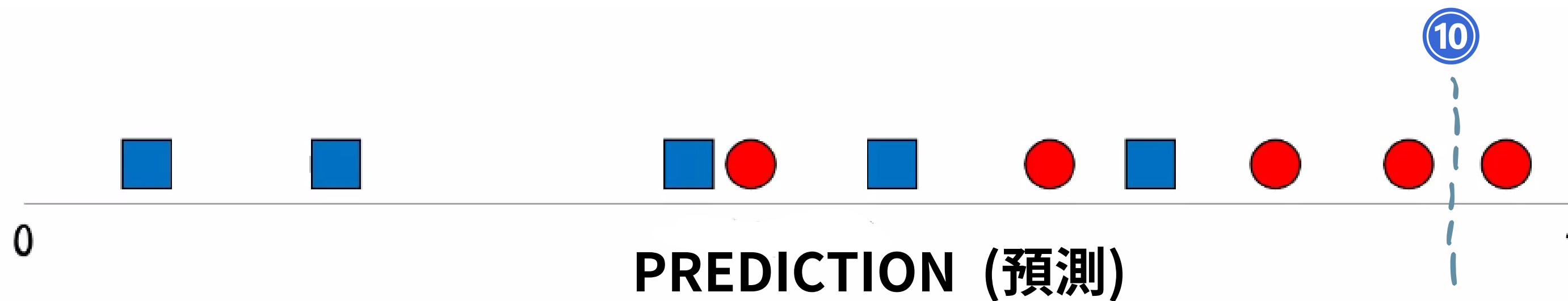
	真實值	
預測值	2	0
	3	5

$$TPR = TP / (TP + FN)$$

$$= 2 / (2 + 3) = 0.4$$

$$FPR = FP / (FP + TN)$$

$$= 0 / (0 + 5) = 0$$



● Positive
■ Negative

TRUTH (真實)

 Cut-off(閾值)

10 閾值

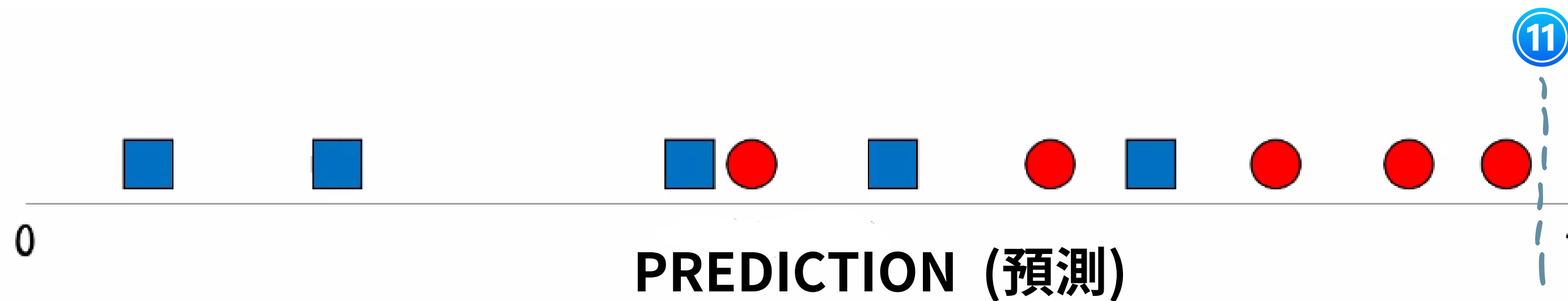
	真實值	
預測值	1	0
	4	5

$$TPR = TP / (TP + FN)$$

$$= 1 / (1 + 4) = 0.2$$

$$FPR = FP / (FP + TN)$$

$$= 0 / (0 + 5) = 0$$



11 閾值

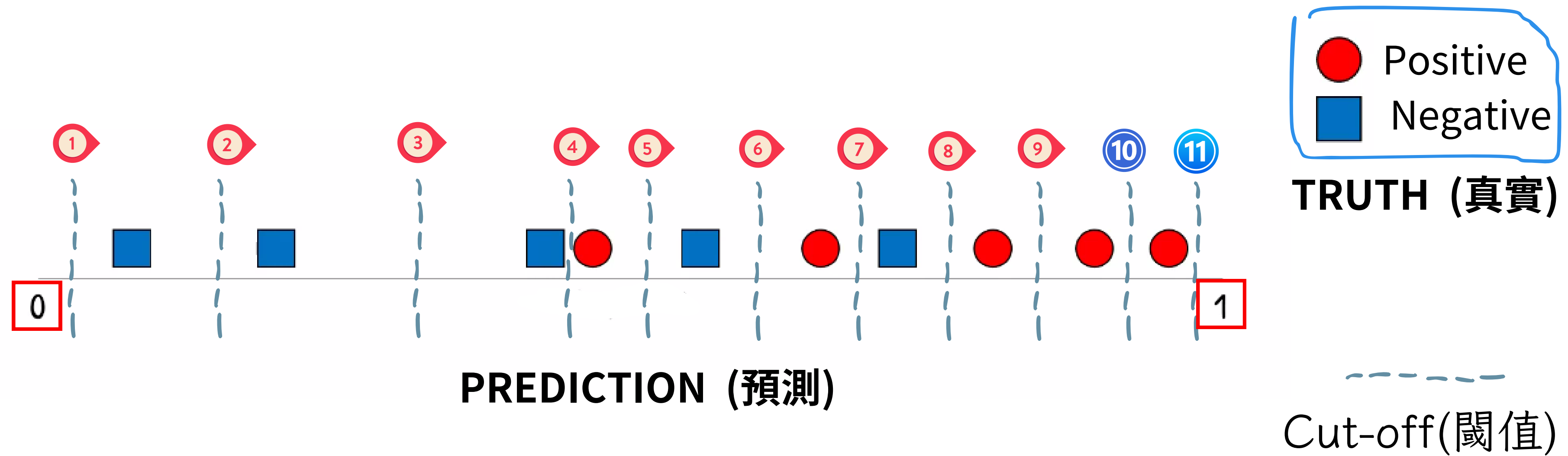
	真實值	
預測值	0	0
	5	5

$$TPR = TP / (TP + FN)$$

$$= 0 / (0 + 5) = 0$$

$$FPR = FP / (FP + TN)$$

$$= 0 / (0 + 5) = 0$$



FPR (x)	0	0	0	0	0.2	0.2	0.4	0.4	0.6	0.8	1
TPR (y)	0	0.2	0.4	0.6	0.6	0.8	0.8	1	1	1	1

FPR (x)	0	0	0	0	0.2	0.2	0.4	0.4	0.6	0.8	1
TPR (y)	0	0.2	0.4	0.6	0.6	0.8	0.8	1	1	1	1

